

Present and Ready: A Liveness Detection System for Real-Time Face Recognition Attendance Monitoring

Rhodmin Lou B. Berioso
Department of Information
Technology, College of
Computing Studies, WMSU,
rhodmin13@gmail.com

Guilmar A. Quimba
Department of Information
Technology, College of
Computing Studies, WMSU,
qguilmar@gmail.com

Lawrence Olivier N. Tilde
Department of Information
Technology, College of
Computing Studies, WMSU,
lawrencetilde@gmail.com

Sir Rhamirl B. Jaafar
Department of Information
Technology, College of
Computing Studies, WMSU,
rhamrhemjaafar@gmail.com

Abstract

— Manual and RFID-based attendance methods in academic institutions remain susceptible to proxy fraud, recording errors, and instructional time loss. This paper presents Present and Ready, a real-time classroom attendance system that integrates facial recognition and liveness detection to address these challenges. The system employs YOLOv8 for face and body detection, InsightFace for 512-dimensional facial embedding, TorchReID for identity tracking across frames, and an ONNX-based anti-spoofing model augmented with Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) heuristics for liveness validation. Twelve domain-specific heuristic rules govern the pipeline, covering face presence consistency, liveness-first gating, one-person-one-log enforcement, ambiguity gap checking, and cross-section identity guarding, among others. The system is deployed via a Flask web framework backed by a MySQL database, accessed through RTSP streaming from a Hikvision 4K camera. Evaluation was conducted across 12 class sessions involving 15 participants, yielding 432 face recognition attempts. Results show 85.2% face detection accuracy, 85.0% recognition precision, 89.8% liveness detection accuracy, a 2.1% false acceptance rate, and an average response latency of 10.35 seconds. A one-week pilot deployment achieved an average attendance accuracy of 82.67% across normal, dim, and high-movement conditions. A user evaluation based on ISO/IEC 25010 quality attributes produced an overall satisfaction score of 4.56 out of 5.00 (91.2%). These findings confirm that integrating multi-layer liveness detection with real-time facial recognition provides a secure, automated, and practical approach to classroom attendance monitoring suitable for resource-constrained academic institutions.

Index Terms—Attendance monitoring, facial recognition, liveness detection, anti-spoofing, YOLOv8, InsightFace, TorchReID, deep learning, educational technology, WMSU.

I. INTRODUCTION

Attendance monitoring remains a critical yet inefficient administrative task in Philippine universities. At the College of Computing Studies (CCS) of Western Mindanao State University (WMSU), roll calls consume two to eight minutes per session, accounting for one to five percent of total instructional hours across a semester. Beyond time loss, paper-based records are vulnerable to loss, unauthorized alteration, and proxy attendance, where absent students are fraudulently marked present. These problems directly undermine academic accountability and data integrity [1].

Facial recognition has emerged as a compelling solution owing to its non-intrusiveness and scalability. However, most existing systems lack robust presentation attack detection (PAD), leaving them exploitable via printed photographs, screen replays, or video spoofs. Several local systems have demonstrated feasibility but remain incomplete in their defenses. Reynoso and Torres [10] developed Tempus, a cloud-based recognition system that improved record accessibility yet lacked anti-spoofing entirely. Grefaldo and Bausa [11] achieved automated logging at a Philippine high school but similarly omitted liveness detection. AttenFace [12] reduced computational overhead using periodic snapshots without any PAD layer. Internationally, Demir and Savas [13] and Abderraouf et al. [14] explored web-backend and embedded architectures, while Motwani et al. [15] scaled pipelines to large venues, yet none evaluated spoof resistance in an operational classroom. On the technology side, InsightFace [18] provides high-accuracy 512-dimensional embeddings, YOLOv8 [19] enables real-time detection on lightweight hardware, and deep anti-spoofing frameworks [9][22][23] have shown strong performance on benchmark datasets. The consistent gap across this body of work is the absence of end-to-end, liveness-verified attendance systems validated in a real classroom environment.

This paper addresses that gap by presenting Present and Ready, a real-time attendance system that combines YOLOv8, InsightFace, TorchReID, and an ONNX-based liveness model into a unified pipeline reinforced by twelve domain-specific heuristic rules. The system was deployed and evaluated in an actual WMSU-CCS classroom. Its primary contributions are: (1) a complete, liveness-integrated attendance pipeline operable on consumer hardware; (2) a twelve-rule heuristic layer covering ambiguity gap enforcement and cross-section identity guarding; (3) empirical evaluation across 432 recognition attempts in a live Philippine university classroom; and (4) a user satisfaction assessment using ISO/IEC 25010 quality attributes.

II. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture

The system follows a client-server model. A Hikvision DS-2CD2086G2-I(U) 4K AcuSense camera streams live video via RTSP to a Lenovo LOQ 15IRX9 laptop hosting the Flask web application. Attendance records are stored in a cloud-based MySQL database accessible through any web browser. The central per-frame function, `refresh_with_detections()`, applies a frame-skip factor (`SKIP_FACTOR = 2`) to process every other frame, reducing CPU load while keeping track states current on skipped frames. The pipeline is illustrated in Fig. 1.

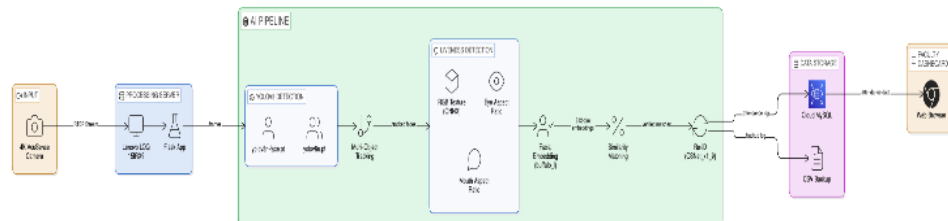


Fig. 1. Present and Ready pipeline: RTSP Camera to YOLOv8 Detection, ByteTrack Tracking, Liveness (ONNX+EAR/MAR), InsightFace Embedding, Cosine Matching, and Attendance Logging (MySQL+CSV).

B. AI Models and Machine Learning Pipeline

1) Face and Body Detection (YOLOv8): The system uses two YOLOv8 models: yolov8n-face.pt for facial region detection and yolov8n.pt for full-body detection, selected for real-time inference speed and bounding-box accuracy on lightweight hardware [19]. Detected bounding boxes are passed to ByteTrack for persistent cross-frame tracking with stable IDs.

2) Facial Recognition (InsightFace): InsightFace (buffalo_l) generates L2-normalized 512-dimensional embeddings for each liveness-verified face. For profile faces, detected via facial-landmark keypoints, a supplementary multi-angle encoding pool is used. Similarity is computed as $S(f_1, f_2) = (f_1 \cdot f_2) / (|f_1| \cdot |f_2|)$. A match requires: (a) similarity within the adaptive threshold 0.57–0.63 for frontal faces or lower for profiles; and (b) an ambiguity gap of ≥ 0.12 over the second-best candidate. Attendance is logged only after CONFIRMATION_FRAMES_REQUIRED consecutive confirmed frames, preventing single-frame false positives [18].

3) Identity Tracking (TorchReID): TorchReID OSNet_x1_0 blends per-frame ReID vectors via exponential moving average ($\alpha = 0.90\text{--}0.95$). Updates pass four sequential validation stages: spatial proximity, body-signature consistency (ReID distance ≤ 0.25), temporal consistency, and IoU overlap. A cross-section guard rejects faces from other enrolled sections, logging them as 'Unknown – Wrong Section' [31].

4) Liveness Detection (ONNX + Facial Landmarks): Liveness verification combines three mechanisms: (a) an ONNX-based RGB texture classifier (model_rgb.onnx) using Laplacian analysis to reject prints and replays; (b) Eye Aspect Ratio (EAR) for blink detection; and (c) Mouth Aspect Ratio (MAR) for mouth movement. All three checks must pass before recognition is attempted [30].

C. Heuristic Rule System

Twelve domain-specific heuristic rules (Table I) are layered over the deep learning pipeline to enforce operational safeguards, governing detection consistency, liveness gating, confidence thresholds, identity deduplication, spatial association, session bounds, fail-safe logging, ambiguity gap enforcement, and cross-section identity guarding.

TABLE I HEURISTIC RULES APPLIED IN ATTENDANCE DECISION-MAKING

Rule	Description
R1	Face Presence Consistency – stable detection required before recognition
R2	Liveness-First Validation – recognition only after PAD confirmation
R3	EAR/MAR Motion Threshold – physiological motion cues required
R4	Confidence Threshold – adaptive cosine similarity 0.57–0.63 required
R5	One-Person-One-Log – single attendance entry per session per identity
R6	Face-Body Association – face must spatially align with detected body
R7	Unknown Identity Isolation – unmatched faces flagged and excluded
R8	Temporal Attendance Validation – status by timestamp vs. session start
R9	Session-Bound Processing – detection active only during faculty sessions
R10	Fail-Safe Logging – redundant storage in MySQL and CSV backup
R11	Ambiguity Gap Enforcement – top match must exceed second-best by ≥ 0.12
R12	Cross-Section Guard – wrong-section faces logged as 'Unknown – Wrong Section'

D. Software Stack and Ethical Compliance

The backend uses Python 3.12, Flask 3.1.2, OpenCV 4.12, and MySQL 9.3. Security features include bcrypt password hashing, OTP-based email verification, role-based access control, and Cloudflare Tunnel for remote access. The study strictly adhered to the Data Privacy Act of 2012 (R.A. 10173). All participants provided informed consent; facial data is stored only as numerical embeddings. Institutional ethical clearance was obtained prior to deployment.

III. EXPERIMENTAL EVALUATION

A. Functional Testing

All six system modules—face detection, liveness detection, facial recognition, attendance recording, OTP authentication, and the admin dashboard—were tested independently and in integration via a live RTSP feed in both a controlled laboratory and an actual CCS classroom. All modules passed their expected functional criteria with no critical errors or interruptions observed.

B. System Performance Metrics

Evaluation spanned 12 class sessions with 15 participants, producing 432 face recognition attempts. Results are presented in Table II. The 85.2% detection accuracy and 85.0% recognition precision met the $\geq 85\%$ targets. Liveness accuracy of 89.8% narrowly fell short of the 90% goal, attributed to dim-lighting variation. The 10.8 FPS reflects the SKIP_FACTOR = 2 design—processing every other frame halves raw throughput but sustains sufficient responsiveness for 10–15 students. Confidence decay ($\times 0.85$ per frame) prunes stale tracks after 30 frames, stabilizing performance. The 10.35-second response latency falls within acceptable bounds.

TABLE II SYSTEM PERFORMANCE EVALUATION METRICS

Metric	Result
Face Detection Accuracy	85.2%

Face Recognition Precision	85.0%
Liveness Detection Accuracy	89.8%
False Rejection Rate (FRR)	3.4%
False Acceptance Rate (FAR)	2.1%
Average Frame Rate	10.8 FPS
Average Response Latency	10.35 sec

C. Liveness and Anti-Spoofing Evaluation

The liveness module was stress-tested with printed photographs, screen replays, and pre-recorded video. Only 2.1% of spoof attempts were incorrectly accepted (FAR = 2.1%), demonstrating effective multi-layer defense. The ONNX texture classifier rejected the majority of print and replay attacks, while EAR and MAR checks provided secondary physiological verification, consistent with findings in [22].

D. Pilot Attendance Accuracy

Table III reports attendance accuracy under three environmental conditions. Normal lighting yielded 96.8% accuracy, validating the pipeline for standard deployment. Accuracy fell to 70.5% under dim lighting, consistent with known sensitivity of vision-based systems to low illumination [15]. High-movement conditions produced 80.7%, reflecting ByteTrack's stability under moderate motion. The 82.67% average demonstrates classroom readiness, with lighting control as the primary improvement target.

TABLE III ATTENDANCE ACCURACY UNDER DIFFERENT CONDITIONS

Condition	Accuracy (%)
Normal Lighting	96.8%
Dim Lighting	70.5%
High Movement	80.7%
Average Accuracy	82.67%

E. User Satisfaction Evaluation

Faculty and students evaluated the system using a Likert-scale questionnaire (1–5) across three ISO/IEC 25010 attributes (Table IV). The overall satisfaction score of 4.56/5.00 (91.2%) exceeded the 80% target. Functionality scored highest (4.70), reflecting confidence in the recognition and liveness modules. Usability (4.60) confirmed interface intuitiveness. Reliability (4.38) was lowest, with faculty citing tracking instability during occlusion, consistent with the quantitative findings.

TABLE IV USER EVALUATION RESULTS BASED ON ISO/IEC 25010

Quality Attribute	Faculty	Student	Avg
Functionality	4.40	5.00	4.70
Usability	4.20	5.00	4.60
Reliability	3.90	4.87	4.38
Overall Satisfaction	4.15	4.95	4.56

IV. DISCUSSION

A. Effectiveness of the Integrated Pipeline

The combination of YOLOv8, InsightFace, TorchReID, and the ONNX liveness model provides a balanced trade-off between accuracy and real-time performance on consumer-grade hardware. The heuristic rule system proved essential: without R1 (frame buffering) and R5 (one-person-one-log), transient detections and duplicate entries would have significantly reduced attendance precision. The ambiguity gap rule (R11) prevented identity confusion in crowded scenes, while the cross-section guard (R12) eliminated false positives from students of other enrolled sections. The CONFIRMATION_FRAMES_REQUIRED mechanism ensured that attendance was never recorded on a single-frame match, substantially reducing misidentification events.

The 85% recognition precision is lower than controlled-dataset benchmarks due to real-world factors: variable illumination, partial occlusion, and limited enrollment images per participant [18]. Future improvements include domain-adaptive fine-tuning of the buffalo_1 model on Philippine student demographics.

B. Comparison with Prior Work

Table V compares Present and Ready against related systems. While Tempus [10], AttenFace [12], and Demir and Savas [13] provide real-time facial recognition, none include liveness detection. The proposed system is the only one evaluated in an operational Philippine university classroom with full anti-spoofing capability, directly addressing the gap identified across prior work.

TABLE V BASELINE COMPARISON WITH RELATED ATTENDANCE SYSTEMS

System	Face Recog.	Liveness	Real-Time	Classroom
Tempus [10]	Yes	No	Yes	Yes
AttenFace [12]	Yes	No	Yes	Partial

Demir & Savas [13]	Yes	No	Yes	No
Proposed System	Yes	Yes	Yes	Yes (WMSU)

C. Limitations

The 10.8 FPS rate falls below the 20 FPS target, primarily due to running the full AI pipeline on a single laptop GPU. The attendance accuracy degradation under dim lighting (70.5%) indicates a hardware dependency on adequate illumination. The system scope is bounded to classrooms not exceeding 70 sqm with 15 concurrent students. Detection of advanced 3D mask spoofing or deepfake-based attacks remains an identified vulnerability.

V. CONCLUSION

This paper presented Present and Ready, a real-time facial recognition attendance system with integrated liveness detection, deployed and evaluated at WMSU-CCS. The system achieved 85.2% face detection accuracy, 85.0% recognition precision, 89.8% liveness accuracy, a 2.1% false acceptance rate, and a user satisfaction score of 4.56/5.00. Twelve domain-specific heuristic rules—including ambiguity gap enforcement (R11) and cross-section identity guarding (R12)—provided essential operational safeguards that improved reliability and reduced false attendance events. Pilot deployment across 12 class sessions confirmed both technical feasibility and practical value in a live Philippine university setting.

The system demonstrates that effective, multi-layer liveness detection is achievable on consumer hardware without depth sensors or physiological signal equipment, offering a viable model for resource-constrained academic institutions. Future work will address GPU server deployment to reach the 20 FPS target, adaptive lighting compensation, multi-camera campus-wide expansion, advanced deepfake and 3D mask detection, and longitudinal evaluation across full academic semesters.

ACKNOWLEDGMENT

The authors thank their capstone adviser, Mr. Rhamir B. Jaafar, panel members Mr. Salimar Bendanillo Tahil, Mr. Elvin Rey Saavedra, Mr. Ceed Jezreel B. Lorenzo, MIT, and Asst. Prof. Marjorie A. Rojas, and department head Mr. Jason A. Catadman, MIT, for their guidance and support. Special thanks to TechBytes Store and all student volunteers who participated in testing.

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